

# **Detailed Design Report**

**Project: Smart Child Monitoring & Comfort Control System**

**Course: ECE 5000**

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**List of Symbols**

V – Voltage in volts  
A – Current in amperes  
W – Power in watts  
 $\Omega$  – Resistance in ohms  
 $\mu\text{F}$  – Capacitance in microfarads  
lux – Unit of light intensity  
 $^{\circ}\text{C}$  – Degree Celsius  
% – Percentage

**List of Acronyms**

ADC – Analog-to-Digital Converter  
AWG – American Wire Gauge  
BOM – Bill of Materials  
CAM – Camera  
DC – Direct Current  
ESP32-S3 – Espressif 32-bit Microcontroller, S3 Series  
I2C – Inter-Integrated Circuit Communication Protocol  
LM2596 – DC-DC Buck Converter IC/Module  
MOSFET – Metal-Oxide-Semiconductor Field-Effect Transistor  
PCB – Printed Circuit Board  
PWM – Pulse Width Modulation

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# Detailed Design

## System Design

The Smart Child Monitoring & Comfort Control System is designed to provide caregivers with a safe, low-voltage method for monitoring a child's room and responding when conditions in the surrounding area become uncomfortable or unsafe. The system continuously monitors temperature, humidity, light intensity, and sound level. Based on the recorded inputs, the microcontroller decides whether the room is in a normal state or an alert should be triggered. In addition to monitoring, the system allows the caregiver to remotely improve the child's comfort by controlling a 12V fan and a 12V LED light via a web dashboard. The design also includes an ESP32-S3 CAM, allowing camera functionality to be explored as part of the monitoring concept. The overall objective of this project is to integrate sensing, alert generation, active control, and actuation into a single prototype.

The system is controlled by the Freenove ESP32S3-CAM board, which serves as the main controller. [1] A 12V DC adapter supplies 12V DC voltage to the prototype through a barrel jack. The incoming 12V line is protected by a 3A inline fuse, and then it is divided into two paths. One supplies 12V to the comforting devices, such as a DC fan and a strip LED, through a MOSFET driver module. The second path is stepped down to 5V using an LM2596 buck converter to power the controller and sensors. [3] The sensing subsystem includes a DHT11 temperature and humidity sensor, a BH1750 light sensor, and an analog sound sensor. These modules provide the controller with real-time data, which is processed by firmware to compare against programmed thresholds and determine the appropriate system response.

The design originally included camera-based monitoring through the ESP32-S3 CAM board. During testing, the camera operated correctly when the overall system load was relatively low. Moreover, when more subsystems and loads are added, the camera operation becomes unstable. This suggested that the available current or power under high operating load is insufficient for the camera to work reliably. As a result, the camera remained part of the overall design and hardware platform, but it was not included as a fully validated feature in the final prototype. Instead, the final demonstration state focuses on stable sensing, alerting, and output control features. This output reflects a realistic engineering decision based on prototype testing. The Design Level 1 structure of the system consists of five main subsystems: the power and regulation subsystem, the sensor subsystem, the controller and decision subsystem, the output actuation subsystem, and the user monitoring subsystem. The major signals between these subsystems include the regulated 5V rail, sensor data signals, analog sound input, fan and light signals, and alert output signal.

Overall, the system integrates protected power conversion, environmental sensing, alert generation, and remote actuation control into a single child-monitoring prototype. Although the

camera feature is not validated under the final loading condition, the prototype successfully demonstrates the design's core monitoring and comfort control objectives.

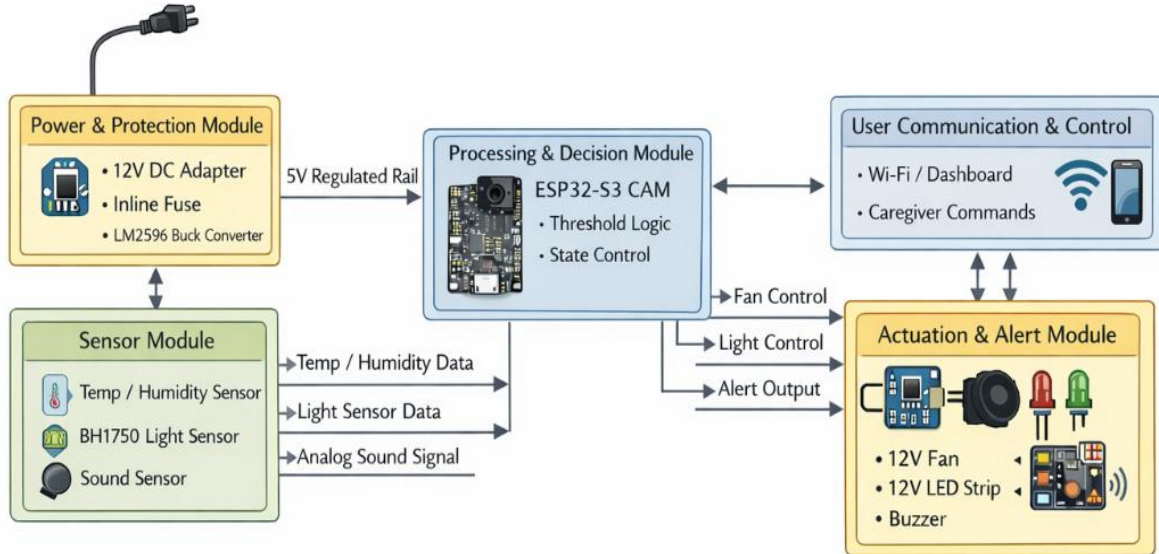


Figure 1: System-level Block Diagram

## System Specification

Table 1: Electrical Specifications

| Parameter                | Description                                                                     |
|--------------------------|---------------------------------------------------------------------------------|
| Input supply             | 12V DC adapter                                                                  |
| Input current capability | 1.5A adapter used in design baseline                                            |
| Protection               | Inline fuse on 12V input                                                        |
| 5V rail generation       | LM2596 buck convertor, 12V to 5V [3]                                            |
| Main controller          | Freenove ESP32S3-CAM [2]                                                        |
| Sensors                  | Temperature & Humidity Sensor, BH1750 light sensor, analog sound sensor [4] [5] |
| Controlled loads         | 12V DC fan, 12V LED strip                                                       |
| Load switching           | 2 MOSFET driver modules                                                         |
| Indicators               | Status LED, alert LED & active buzzer                                           |
| Connector type           | DC barrel jack and connection wires                                             |

Table 2: Mechanical Specifications

| Parameter           | Description                        |
|---------------------|------------------------------------|
| Enclosure type      | 3D printed enclosure               |
| Enclosure material  | Polylactic Acid (PLA)              |
| Overall dimensions  | 120mm*120mm*8.85mm                 |
| Total system weight | 800g                               |
| Mounting Method     | Internal Standoffs, Nuts and Bolts |

|                       |                                                                                  |
|-----------------------|----------------------------------------------------------------------------------|
| Sensor access         | Openings for light sensor, sound input, temperature & humidity sensor & two LEDs |
| External access       | Power Input, fan & light wiring exits and visible indicators                     |
| Cooling               | Sensor Outlets acts as vents for cooling                                         |
| Operating environment | At room temperature for child's room                                             |

## Circuit Design & Calculation

### Power Subsystem

The power subsystem is designed to operate safely with both low voltage electronics and high current 12V output loads from a single DC source. A 12V adapter is connected to the system through a soldered DC barrel jack on the perfboard. An inline fuse is placed between the barrel output and 12V rail to provide protection in the case of wiring faults, short circuits or unexpected load conditions. After protection the rail is divided into two branches. The first branch supplies the power to the 12V DC fan and 12V LED strip through two MOSFET driver modules. The second branch feeds a LM2596 buck convertor that steps down 12V input to 5V for the ESP32-S3 CAM board and sensor modules. Bulk capacitors and decoupling capacitors are used on the supply rails to improve stability during switching loads. This arrangement provides a satisfactory design for voltage architecture while maintaining electrical isolation between different subsystems of the prototype.

This architecture is selected because the fan and LED strip are 12V DC loads while the controller and sensor require a lower regulated voltage. Using buck convertor instead of a linear regulator improves the efficiency and reduces unnecessary wastage of power. The use of both buck convertor and small valued decoupling capacitors helps the system to tolerate rapid changes in current caused by load switching and wireless control activity.

### Sensor Interfacing Subsystem

The sensing subsystem provides the controller with all the environmental data needed to assess the atmosphere of the child's room. A temperature and humidity sensor is used to monitor thermal comfort conditions. A BH1750 module is used to measure the ambient light intensity in lux. [5] An analog sound sensor module is used to detect the sustained crying events. The digital sensors communicate with the ESP32-S3 through I2C communication pins, and the sound sensor provides an analog signal that is detected by the ADC pin. The firmware loaded into the controller periodically samples these sensor values, interprets them and compares them with programmed thresholds to determine whether the environmental conditions remain within the acceptable limits.

The sensor selection supports the project's needs because it combines room comfort measurement with event detection. Temperature and humidity reflect air comfort, light sensing can detect excessive bright conditions, and sound sensing provides a practical indication a

sustained cry event. Altogether, these measurements allow the controller to make decisions that are more meaningful as compared to a single sensor system.

### Output Actuation Subsystem

The output subsystem consists of a 12V DC fan, a 12V LED strip, indicator LEDs and an active buzzer. The 12V fan and LED strip are not directly powered from the ESP32-S3 CAM. [2] Instead, they are powered through the 12V rail through the MOSFET driver modules, which allows the segregation of low voltage signals from high current loads to ensure a safe operation. This approach improves reliability and protects the microcontroller from increased current demand. The buzzer and status LEDs are used to provide feedback during the normal operation, alert conditions and warning conditions.

### Control and Decision Logic

The controller firmware continuously reads sensor values and evaluates them against programmed thresholds. When all the sensor conditions remain within the acceptable values the system stays in a normal operation state. When the measured values exceed the set limit, the firmware state changes and activates the alert indication then the dashboard is updated. In addition to the automatic response, the caregiver can remotely control comfort controls such as the fan and light. This makes the system both responsive and user assisted which is an important aspect for designing a child monitoring system. The project background also shows the remote control of the 12V fan and 12V LED strip as one of the key functions of the design.

The ESP32-S3 CAM board is selected as the microcontroller because it combines the process capability with communication features and integrated camera hardware. The camera capability was part of the original system design, since visual verification during the alert event be a useful feature added to the prototype. However, testing showed that the camera could operate only under light load conditions, while when load is increased the camera becomes unreliable. This suggested that the available current or voltage is insufficient to support the use of the camera during a fully active system. Therefore, the final prototype is mainly focused on stable sensing, alerting, dashboard updating and comfort control.

### Design Calculations

To explain the choice of components to design this prototype, the following are some calculations.

#### LED Current Limiting Resistor Calculation

The indicator LEDs use 220  $\Omega$  series resistor. Assuming a 5V supply rail and a LED forward voltage of 2V, the LED current can be estimated using Ohm's law:



$$I = \frac{V_{supply} - V_f}{R}$$

$$I = \frac{5-2}{220} = \frac{3}{220} = 0.0136 \text{ A}$$

$$I \approx 13.6 \text{ mA}$$

Therefore, the selected 220  $\Omega$  resistor limits LED current to 13.6 mA which is suitable for visible status indication while remaining within a safe range for a standard signal LED.

### Input Power Capability Calculation

The prototype uses a 12V adapter in 1.5A design range. The available input power is given by

$$P = 12 \times 1.5 = 18\text{W}$$

Thus, the available input power is close to 18W. This supports the needs of the design for operating both the logic electronics and the 12V fan and LED strip loads from a single power source.

### Buck Conversion Ratio

The LM2596 buck convertor steps down the supply voltage from 12V to 5V. The voltage conversion ratio is

$$\frac{V_{out}}{V_{in}} = \frac{5}{12} = 0.417$$

Therefore, the convertor output is approximately 41.7% of the input voltage, which provides a suitable supply voltage for controller and low voltage sensors.

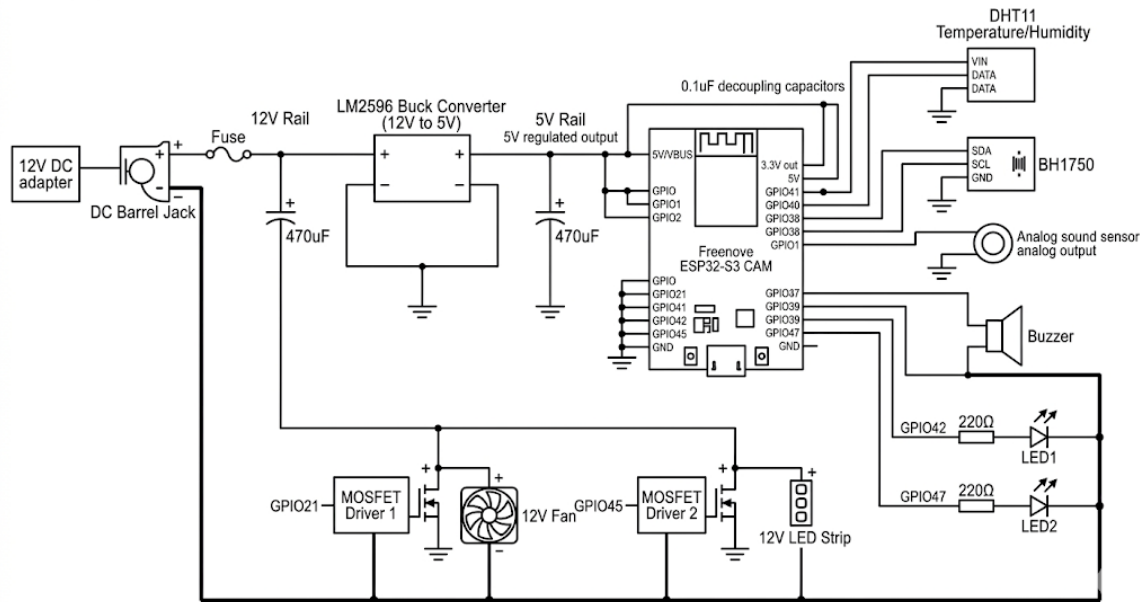


Figure 2: Complete Circuit Schematic

## Bill of Materials

Table. 3 presents the bill of materials for the smart child monitoring and comfort control system. The BOM includes the controller hardware, power supply modules, sensing modules, output devices and indication components required to assemble the final prototype.

Table 3: Bill of Materials

| Component Name                                | Quantity | Unit Price | Total Price | Purchasing Link                                                                                                                                                                          |
|-----------------------------------------------|----------|------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Freenove ESP32-S3 CAM board                   | 1        | C\$ 31.95  | C\$ 31.95   | <a href="#">FREENOVE ESP32-S3-WROOM Powerful Wireless Camera Board</a>                                                                                                                   |
| 12V DC Wall Adapter (1.5A rating)             | 1        | C\$ 15.99  | C\$ 15.99   | Lab Sourced                                                                                                                                                                              |
| LM2596 Buck Convertor Module (12V to 5V)      | 1        | C\$ 2.60   | C\$ 2.60    | <a href="#">DDUAN 5-Pack LM2596 DC-DC Adjustable Step Down Voltage Regulators</a>                                                                                                        |
| DC Barrel Jack, Panel Mounted, 5.5mm X 2.1mm  | 1        | C\$ 2.33   | C\$ 2.33    | Lab Sourced                                                                                                                                                                              |
| Inline fuse                                   | 1        | C\$ 3.40   | C\$ 3.40    | Lab Sourced                                                                                                                                                                              |
| 470 uF Electrolytic Capacitors, 25V or higher | 2        | C\$ 0.87   | C\$ 1.74    | Lab Sourced                                                                                                                                                                              |
| 0.1 uF Ceramic Capacitor                      | 5        | C\$ 0.1    | C\$ 0.5     | Lab Sourced                                                                                                                                                                              |
| DHT11 Temperature and Humidity Sensor Module  | 1        | C\$ 9.99   | C\$ 9.99    | <a href="#">Organizer 5 Pcs DHT11 Temperature and Humidity Sensor Module for Arduino UNO MEGA 2560 AVR PIC Raspberry Pi 2 3 4B (5pcs DHT11) : Amazon.ca: Industrial &amp; Scientific</a> |
| BH1750 Light Sensor Module                    | 1        | C\$ 11.46  | C\$ 11.46   | <a href="#">BH1750 Module, GY-302 Digital Light Sensor BH1750 Chip Integration Module 3-5V Power Supply : Amazon.ca: Industrial &amp; Scientific</a>                                     |
| Analog Sound Sensor Module                    | 1        | C\$ 3.62   | C\$ 3.62    | Lab Sourced                                                                                                                                                                              |
| 12V DC Fan                                    | 1        | C\$ 11.00  | C\$ 11.00   | Lab Sourced                                                                                                                                                                              |
| 12V LED Strip                                 | 1        | C\$ 11.00  | C\$ 11.00   | Lab Sourced                                                                                                                                                                              |

|                             |                  |           |           |                                                                                                                                               |
|-----------------------------|------------------|-----------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| MOSFET Driver Module        | 2                | C\$ 3.91  | C\$ 7.82  | <a href="#">DAOKI 10pcs 15A 400W MOS FET Trigger Switch Drive Module PWM Regulator Control Panel : Amazon.ca: Industrial &amp; Scientific</a> |
| Active Buzzer Module        | 1                | C\$ 2.91  | C\$ 2.91  | Lab Sourced                                                                                                                                   |
| Indicator LEDs              | 2                | C\$ 0.09  | C\$ 0.18  | Lab Sourced                                                                                                                                   |
| 220 Ohm Resistors           | 2                | C\$ 0.10  | C\$ 0.20  | Lab Sourced                                                                                                                                   |
| Perfboard                   | 1                | C\$ 1.37  | C\$ 1.37  | Lab Sourced                                                                                                                                   |
| 2-pin Screw Terminal Blocks | 3                | C\$ 0.22  | C\$ 0.66  | Lab Sourced                                                                                                                                   |
| 20 AWG Wires                | 5 meter (Approx) | C\$ 2.00  | C\$ 2.00  | Lab Sourced                                                                                                                                   |
| Heat Shrink                 | 1 meter          | C\$ 1.00  | C\$ 1.00  | Lab Sourced                                                                                                                                   |
| Zip Ties                    | 10               | C\$ 0.10  | C\$ 1.00  | Lab Sourced                                                                                                                                   |
| M3 Screws & Standoffs       | 4 each           | C\$ 2.00  | C\$ 2.00  | Lab Sourced                                                                                                                                   |
| 3D Printed Enclosure        | 1                | C\$ 10.00 | C\$ 10.00 | Lab Sourced                                                                                                                                   |

The bill of materials is organized to reflect the final prototype configuration, including sensing, control, power, actuation, indication and packaging hardware. Items sourced from the laboratory inventory were identified as lab stock, while purchased components are shown using the exact supplier links along with actual unit prices. This format provides a complete record of the hardware required to reproduce the prototype and support future redesign, costing and marketing work.

## Mechanical and Package Details

The final prototype is designed with an enclosure to make the system safer, cleaner and easier to use. Since the project includes a 12V power supply, wiring, sensors, output devices, and a microcontroller, leaving all the parts exposed would make the prototype look unfinished and increase the chance of wires getting disconnected or touched accidentally. For this reason, the product is designed to be mounted inside a 3D printed enclosure.

The main purpose of the enclosure is to protect the electronics while keeping the important components accessible from outside. The system is built around an ESP32-S3 CAM board, a 12V DC input, a buck convertor, sensor modules, a buzzer, indicator LEDs, MOSFET driver modules, a 12V fan and a 12V LED strip. Since so many parts are used in this project, the enclosure helps to keep the internal wires organized. The use of perfboard, terminals, heat shrinks, standoffs and a 3D printed enclosure supports this goal of creating a complete and professional prototype. [1]

The enclosure layout is planned so that each sensor can do its job properly. The temperature and humidity sensor, DHT11, needs exposure to room air, so it cannot be sealed inside a box. [4] The BH1750 light sensor needs a clear opening so that it can detect the light level in the room. [5] The sound sensor also needs an opening so that sound in the room can reach it safely. In the same way, the LED indicators must remain visible from outside the enclosure, since there are already some openings in the enclosure so buzzer can be placed near those openings so that it can be heard clearly. This means the package design is not only concerned about looks but also about making sure the system could still monitor the room, correctly after assembly.

Special attention is also given to the power and wiring side of the enclosure. The system uses a 12V adapter, DC barrel jack, inline fuse, LM2596 buck convertor and output wiring for the fan and the LED strip. These parts need support and enough internal space so that wires are not bent or pressed against other components. The enclosure therefore needs openings for DC power input and for the wires going to fan and LED strip. Inside the enclosure, the parts are positioned in a neat way so that the power section, controller section and sensor sections do not interfere with each other. Keeping the layout neat also makes troubleshooting easier if any part needs to be checked later.

Another important reason for using the enclosure is overall presentation. The project is not only meant to work electrically but also to appear as a completely designed product. A 3D printed case gives the prototype a more finished look and makes it easier to present in a professional manner. It also helps show the design is far from a simple breadboard design and towards a real packaged product. In this project, the enclosure supports the goal of safe child monitoring with remote comfort control through the fan and light outputs.

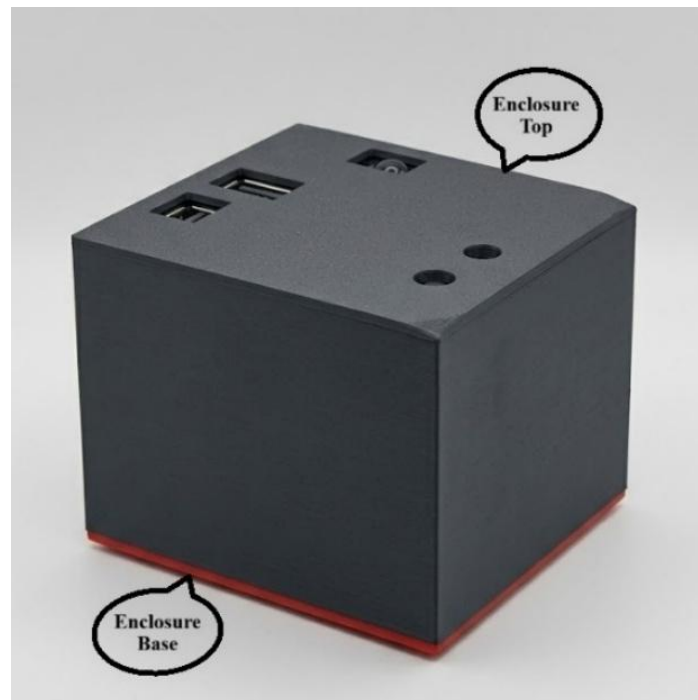


Figure 3: Isometric View of the Prototype

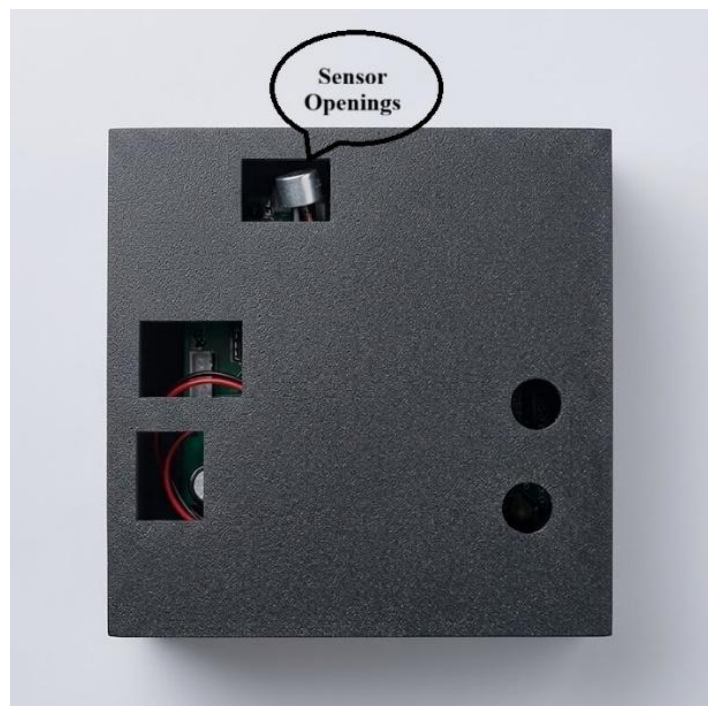


Figure 4: Top View of the Enclosure Showing Sensor Openings

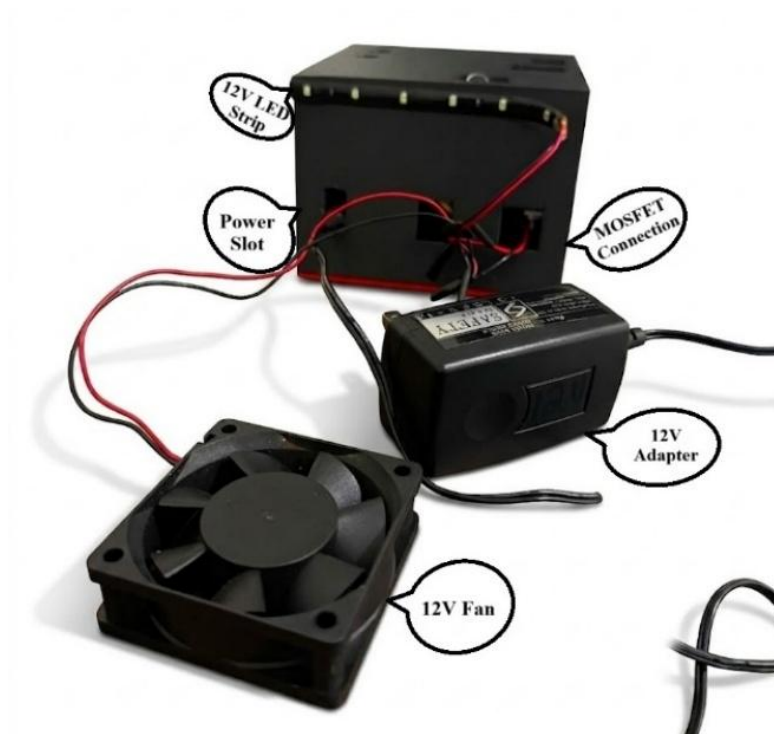


Figure 5: Side View of the Enclosure Showing Power Entry



Figure 6: Inside Circuitry of the Enclosure



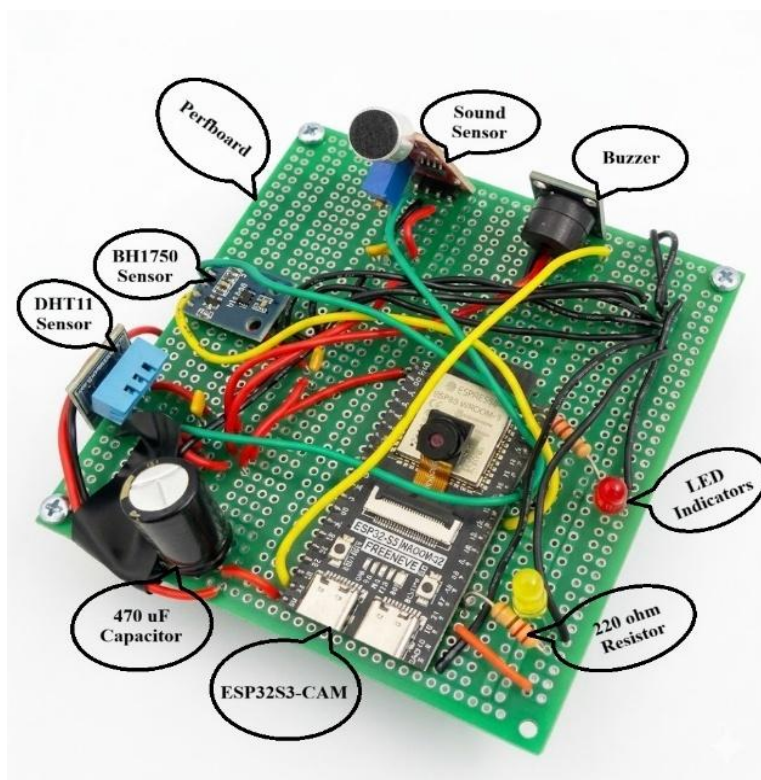


Figure 7: Close-up View of the Sensor Integration

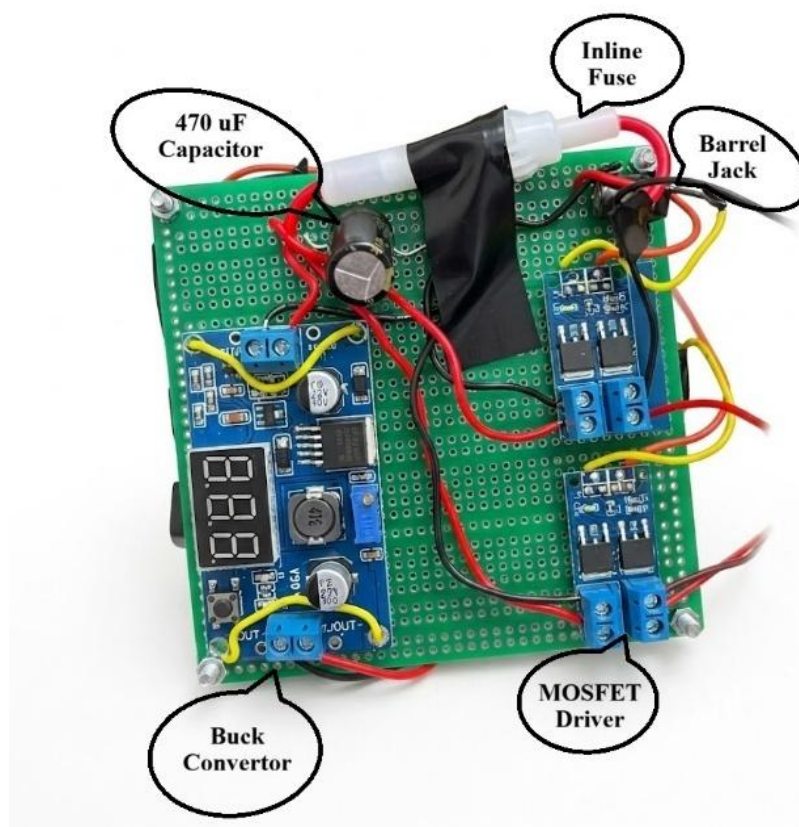


Figure 8: Close-up view of the Power Conversion Module

# Design Verification and Testing

## Acceptance Testing

Acceptance testing is carried out on the final prototype to confirm that the system is working properly under normal user conditions. In this project, the acceptance test is used to check whether the system can take the power safely, read the room conditions correctly, detect the unsafe or uncomfortable conditions, give an alert, and allow the caregiver to control the fan and light. Since acceptance testing is used here to validate the final prototype as a whole, the tests are written in a simple step by step manner. All the tests show that the product meets its expected purpose as all of them are clear, traceable, and verifiable.

The final prototype for this project is built around an ESP32-S3 CAM board, a protected 12V power input, an LM2596 buck convertor, sensors, a buzzer, LEDs, a 12V fan and a 12V LED strip. The key functions of the system are room monitoring, alert generation and remote comfort control. Therefore, the acceptance tests are focused on these final working features.

Table 4: Acceptance Test Cases

| Test Objective                  | Step by Step Test Procedure                                                                                                                                                                                                    | Results                                                                                                      | Comments                                                                                                    | Test Pass/Fail |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|----------------|
| Safe startup of the full system | <ul style="list-style-type: none"> <li>- Connect the 12V adapter to the system</li> <li>- Turn the system on</li> <li>- Observe the controller, LEDs and dashboard</li> </ul>                                                  | The system powers on properly, no abnormal smell is noticed, and dashboard starts showing live data          | System started normally and sensor data appeared within a few seconds                                       | Pass           |
| Temperature & humidity sensing  | <ul style="list-style-type: none"> <li>- Let the system run in normal conditions</li> <li>- Observe the temperature &amp; humidity readings</li> <li>- Breathe near the sensor gently</li> <li>- Watch the readings</li> </ul> | The temperature & humidity reading is changing in a correct manner, and updates can be seen on the dashboard | The change in temperature was reasonable during the test, as it increased when breath was taken near sensor | Pass           |
| Light sensing                   | <ul style="list-style-type: none"> <li>- Observe the normal light reading</li> <li>- Shine a flashlight on the sensor</li> <li>- Cover the sensor</li> </ul>                                                                   | The lux value increases when extra light is applied and decreases when the sensor is covered                 | Lux reading was increased when flashlight was kept on the sensor as expected                                | Pass           |



|                            |                                                                                                                                                                                                                           |                                                                                                            |                                                                                                                  |      |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|------|
| Sound alert function       | <ul style="list-style-type: none"> <li>- Keep the system in normal condition first</li> <li>- Create a sustained sound near the sensor so it reaches above the threshold</li> <li>- Observe the alert response</li> </ul> | The system changes from normal mode to alert mode and activates and buzzer/LED warning and dashboard alert | State of alert was set when high pitched sound was kept on the sound sensor                                      | Pass |
| Bright light action        | <ul style="list-style-type: none"> <li>- Shine strong light on the light sensor until the lux value goes above the threshold</li> <li>- Observe the system response</li> </ul>                                            | System detects the bright light and switches the LED strip in AUTO mode                                    | LED strip was turned off when a flashlight was shined on the light sensor which serves as a feature of AUTO mode | Pass |
| Manual fan control         | <ul style="list-style-type: none"> <li>- Open the dashboard</li> <li>- Send a command to turn the fan ON</li> <li>- Observe the fan</li> <li>- Send the command to turn the fan OFF</li> </ul>                            | The fan turns ON and OFF correctly when commanded                                                          | Command of fan turned ON was served in a few seconds as expected                                                 | Pass |
| Manual light control       | <ul style="list-style-type: none"> <li>- Open the dashboard</li> <li>- Send the command to turn the light ON.</li> <li>- Observe the LED strip</li> <li>- Send the command to turn the light OFF</li> </ul>               | The light turns ON and OFF correctly when commanded                                                        | Command for LED strip to turn ON was seen fulfilled in a few seconds of pressing the button                      | Pass |
| Local indicators           | <ul style="list-style-type: none"> <li>- Trigger an alert using sound</li> <li>- Watch the LED and Buzzer</li> </ul>                                                                                                      | The local indicators clearly show an alert condition has happened                                          | Red LED & the buzzer turned ON when the state of alert was set                                                   | Pass |
| Return to normal condition | <ul style="list-style-type: none"> <li>- Trigger an alert</li> <li>- Remove the abnormal condition</li> </ul>                                                                                                             | The system leaves the alert mode and returns to normal monitoring once the alert is removed                | When the state of alert was removed from the dashboard and the conditions were normal, the                       | Pass |

|                             |                                                                                                           |                                                                                        |                                                                              |      |
|-----------------------------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------|------|
|                             | - Continue observing the system                                                                           | and conditions become normal again                                                     | system turned to the normal state                                            |      |
| Short term stable operation | - Leave the system powered for 10 to 15 minutes<br>- Watch sensor readings, dashboard updates and outputs | The system keeps running without freezing, unexpected reset or random output switching | System was left ON and idle for 15 mins and it performed as normal afterward | Pass |

The acceptance tests shows that the final prototype has performed well as a complete system. The most important things that are checked are safe startup, correct sensor reading, proper alert generation, manual control of fan and light and stable return to normal operation. These tests are chosen because they directly match the main purpose of the project, which is to monitor the child's room, warn the caregiver when conditions become unsafe and provide simple comfort control through the fan and light.

## Integration Testing

For this project, the selection of integration test is the integration between the Sensor Module and the Processing & Decision Module. This integration is chosen because it is an important connection in the system. The temperature and humidity sensor, the light sensor and the sound sensor all send information to the controller. The controller should then read and process that information, compare it with threshold limits and then decide if the system should stay in the normal mode or move to the alert mode. If this connection does not work properly, the full child monitoring system cannot perform all the necessary features.

### Purpose of this test

For this test, the Sensor Module subsystem and the Processing & Decision Module subsystem are chosen. The purpose of this test is to verify that the controller could correctly receive sensor value in real time from the sensor module, process these values, compare them with the set thresholds and change the system state accordingly. This confirms that the sensing and the logic of the design are working perfectly together as a connected system instead of as separate parts.

### Test Setup

The prototype was powered by a 12V adapter and whole system was powered through a single source. The dashboard was used to observe the controller response. Light changes were shown using a flashlight, sound events were created near the sound sensor and slight temperature or humidity changes were created by breathing near the sensor. This setup matches the project hardware and the way the system is intended to work in real life situations.

Table 5: Integration Test Case Table

| Step | Action                                                                                                 | Result                                                                                                        | Comments                                                      | Test Pass/Fail |
|------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|----------------|
| 1    | Connect the 12V adapter and power on the system. Allow the controller and sensor to initialize         | The system started normally and began showing live sensor readings.                                           | Startup was normal and the system initialized properly.       | Pass           |
| 2    | Leave the system under normal room conditions and observe the dashboard                                | The controller received stable sensor data and kept the system in normal mode                                 | The system stayed stable during normal operation              | Pass           |
| 3    | Gently breath near the temperature & humidity sensor                                                   | The controller gets updated temperature and humidity and the displayed reading changes                        | Sensor responded properly to the environmental change         | Pass           |
| 4    | Shine a flashlight on the light sensor                                                                 | The controller received the increased light reading and updated the light value correctly                     | Light sensor reacted clearly to the flashlight                | Pass           |
| 5    | Remove the flashlight and return the sensor to normal room lightning                                   | The controller detected the lower light level and returns the displayed value to the normal range             | Reading dropped after the light was removed                   | Pass           |
| 6    | Create a sustained sound near the sound sensor such that the sound value rises above the set threshold | The controller received the sound signal, processed it correctly and changed the system state to alert mode   | Alert was triggered correctly when sound went above the limit | Pass           |
| 7    | Stop the sound and allow the surrounding to be quite again                                             | The controller detected that the value is dropped and alert is cleared so it returned back to the normal mode | Alert cleared once the sound went back down                   | Pass           |

|    |                                                                   |                                                                                                                |                                                     |      |
|----|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|------|
| 8  | Shine strong light again so the light crosses the alert condition | The controller read the light signal correctly and switches the LED strip OFF when the system was in AUTO mode | High light condition was detected without any issue | Pass |
| 9  | Remove the strong light and continue monitoring                   | The controller returned system to the normal mode when the light sensor value is back in the acceptable range  | System went back to normal state correctly          | Pass |
| 10 | Repeat the high sound condition again                             | Same responses again, as in Step 6, showing that sensor and controller worked together reliably                | Repeated test in Step 6 gave the same result again  | Pass |

This integration test checks whether the Sensor Module and the Processing & Decision Module work together perfectly. It not only checks whether the sensors can produce values but also checks whether the controller can understand those values and respond in the correct way. The test checks the real subsystem interface by using a step-by-step method and exercises the main signal path that matters to the child monitoring system. The selected test also matches the project goals, which include measuring temperature and humidity, measuring light intensity, measuring sound levels, comparing to the safe ranges, and detecting events when room conditions are out of normal ranges.

## Unit Testing

For unit testing, the selected component is the sound sensor input and alert threshold function. This component is chosen because it has one clear feature in the system which is, it reads the sound level and helps the controller decide whether the room is in a normal state or an alert state. In this project, high pitched sound is used as a practical indicator for crying, so this feature is an important part of the child monitoring system.

A matrix test is used for this unit because it is a suitable type of testing when the testing method is same, but the input values change. It matches this case well because the same test can be repeated with different sound values to check whether the component behaves correctly below, at and above the threshold.

The purpose of this unit test is to confirm that the sound sensing path worked properly as a single functional part. In simple terms, the test checks whether low values keep the system in normal

mode and whether higher sound values correctly trigger the alert logic. This test is useful because it checks the sound-based decision part on its own before thinking about the full system.

### Purpose of the Test

For this test, sound sensor input and alert threshold function is chosen. This test is performed to verify that sound sensing unit correctly reads different sound levels and responds based on the programmed threshold.

### Test Setup

The system was powered and allowed to initialize. The sound sensor reading was observed on the web dashboard. Different sound levels were then applied to the sound sensor using different samples of sound.

The following steps were repeated for each case:

1. Power on the system and allow the sensor reading to get stable.
2. Apply the approximate required sound level for the test case.
3. Observe the sound reading and system state.
4. Record whether the response matches the expected output.
5. Reset the condition before the next case.

Table 6: Matrix Unit Test Table

| Applied Sound Level | Expected Result                                                 | Actual Result                                            | Comments                                       | Test Pass/Fail |
|---------------------|-----------------------------------------------------------------|----------------------------------------------------------|------------------------------------------------|----------------|
| 0                   | System should stay in normal mode. No alert should be triggered | System stayed in normal mode and no alert was triggered. | No alert was seen at zero input                | Pass           |
| 20                  | System should stay in normal mode                               | System remained normal at this low sound value           | System stayed normal at low sound level        | Pass           |
| 40                  | System should stay in normal mode                               | No alert was triggered and the reading stayed stable     | Reading was stable and no alert was triggered  | Pass           |
| 60                  | System should stay in normal mode                               | System still remained in normal mode                     | Response was stable and no alert was triggered | Pass           |

|                     |                                                                                                        |                                                                         |                                                        |      |
|---------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------|------|
| 70                  | System should stay in normal mode till value is <71                                                    | No alert was triggered at threshold value                               | This test helped confirm the exact boundary behavior   | Pass |
| 72                  | Alert should be triggered                                                                              | Alert was triggered correctly above the threshold                       | Alert started correctly just above the threshold       | Pass |
| 80                  | Alert should be triggered                                                                              | Alert stayed active as expected                                         | High sound level was detected properly                 | Pass |
| 100                 | Alert should be triggered                                                                              | Strong alert response was observed                                      | Alert response stayed active as expected               | Pass |
| 72, then back to 60 | Alert should turn on above threshold, then alert will clear when the sound level returns to safe range | Alert turned on above the threshold and cleared after the value dropped | System returned to normal once the sound level dropped | Pass |

This unit test is designed to check on small but important part of the project by itself. Instead of testing the full system, it is only focused on the sound input and threshold decision behavior. The matrix format is a good choice because only the sound level changed down the row, while the testing method stayed the same. The actual results match the expected outputs, which shows that the sound-based alert feature of the design is working properly as a functional component.

## Operating Instructions

The following instructions explain how to use the smart child monitoring and comfort control system in a simple and safe manner. These instructions are written for a general user and do not require any technical knowledge. The system is designed to monitor room conditions such as temperature, humidity, light level, sound level and to help the caregiver to control a fan and a light when. The hardware components of the project include 12V power input, an ESP32-S3 CAM control board, sensors, a buzzer, LED indicators, a fan and a 12V LED strip. [2]

### Before Turning the System On

- Place the prototype on a flat and stable surface
- Make sure the enclosure openings for the sensor are not blocked
- Keep the system away from water, liquids, and very dusty areas

- Make sure the fan and light output wires are connected properly
- Check that the power adapter is the correct one for the system
- Make sure the DC power plug is firmly connected before use

## Turning the System On

- Plug the 12V adapter into the power input of the system
- Wait a few seconds for the controller to setup
- Watch for LED indicators or dashboard to confirm that system has started
- Check that live room readings begin to appear on the dashboard

## Normal Use of the System

- Let the system run in the child's room during normal operation
- The system will automatically monitor room temperature, humidity, light level & room sound level
- If the room conditions stay within the normal range, the system will continue running in normal mode
- The user does not need to do anything during normal monitoring except observe the readings if needed

## What Happens During an Alert

- If the room becomes too bright or noisy or any parameter goes outside the programmed safe range, then the system will move into alert mode
- When the alert happens, the buzzer and LED indicator will turn on and dashboard will show the alert message
- The user can check the room conditions when an alert is received
- After the room conditions return to normal, the system should return back to normal monitoring mode

## Using the Fan Control

- Open the dashboard or control interface connected to the system
- Find the fan control option
- Press or select the command to turn the fan ON
- Check that the fan starts running
- Press or select the command to turn the fan OFF when it is no longer needed
- If the fan does not respond, check the power connection and output wiring

## Using the Light Control

- Open the dashboard or control interface
- Find the light control option
- Press or select the command to turn the light ON

- Check that the LED strip turns on
- Press or select the command to turn the light OFF when it is no longer needed
- If the light does not respond, check the wiring and output connection

## Checking Sensor Operation

- To check the temperature & humidity sensor, gently breathe near it and see if the values change
- To check the light sensor, shine a flashlight toward it and see if the light reading increases
- To check the sound sensor, create a clear sound near the sensor and see if the sound reading changes
- These checks help confirm that the system is still reading the room correctly

## After Use

- If the system does not need to run anymore, turn it off
- Unplug the 12 V adapter from the system
- Keep the device where wires will not be pulled or damaged

## Safety Notes

- Do not use the system near water
- Do not touch the inside wiring while the system is powered
- Do not block the sensor openings
- Do not block the fan or ventilation areas
- Do not force the power plug into the connector
- Disconnect power before opening the enclosure or changing wiring
- Use the system only with the correct power adapter

## Simple Troubleshooting Guide

- If the system does not turn on:
  - Check the power adapter
  - Check the DC barrel connection
  - Check the fuse and main wiring
- If the fan does not work:
  - Check the fan wiring
  - Check the MOSFET output connection
  - Try the fan command again
- If the light does not work:
  - Check the LED strip connection
  - Check the light control output
  - Try the command again
- If the sensor readings do not change:



- Make sure the sensor is not blocked
  - Restart the system
  - Test the sensor again
- If the alert does not stop:
  - Check whether the room condition is still outside the normal limit
  - Then allow the reading to return to normal and observe the system again

These operating steps allow a general user to power the system safely, observe room conditions, respond to alerts, and control fan and light when needed. The instructions are written in simple point form so that the system can be used easily without special technical knowledge.

## Conclusion

### Collaborative Learning Experience

This project was one of the most useful learning experiences I have had in electrical engineering because it required me to combine many different skills into one working system. Over the past three months, I have had to learn how to connect sensors, work with power conversion, control higher voltage loads safely, write firmware for decision-making, test the system properly, and think about how the final product should be packaged inside an enclosure. The project started as an idea for a smart child monitoring and comfort control system, but during development it became much more than just connecting parts together. It became a full design process in which each technical decision affected the rest of the system. The main goal of the project was to monitor temperature, humidity, light, and sound, send alerts when conditions became abnormal, and allow the caregiver to control a fan and a light safely using a protected low-voltage design.

One thing that worked well in my learning process was building the project step by step instead of trying to complete everything at once. Testing one part at a time helped me understand the role of each subsystem more clearly. For example, it was easier to first confirm that the sensors were giving reasonable values, then confirm that the control logic was working, and then move on to outputs such as the fan, light, buzzer, and indicators. This gradual approach made debugging easier and helped me understand how the whole system was connected. It also showed me that a design project is not only about getting a final output, but also about understanding how to move from an idea to a reliable prototype.

At the same time, the project also had several challenges. One major challenge was power and integration. On paper, it is easy to think that if each part works individually, the whole system will also work when everything is connected. In practice, that was not always true. Some features behaved differently once more subsystems were active at the same time. The camera feature is a good example. Since the ESP32-S3 CAM board was selected as the controller platform, camera functionality was part of the original concept. However, during testing, the camera did not work properly when more parts of the system were running at the same time. This taught an important

lesson: even if a feature sounds good in theory, it still has to be tested in real conditions before it can be trusted in the final system. Another challenge was arranging everything physically. The wiring, placement of components, and enclosure planning took more time than expected because the system had to work properly while also staying safe and organized.

If I had the chance to start this project again with the same amount of time, I would still choose the same idea because it is practical and meaningful. However, I would manage the project differently. I would plan the power system earlier and better understand how much load and current the full system would need. I would also start proper testing earlier instead of waiting until most of the system was already built. Another thing I would do sooner is plan the enclosure, so that wiring, sensor placement, and packaging would not become last-minute problems. I believe these changes would make both the learning experience and the final project better.

Overall, this project helped me grow from simply using electronic modules to thinking more like a system designer. I learned that successful design is not only about making something work once, but about making it understandable, testable, repeatable, and safe. I also learned that real engineering progress comes from trial, error, and improvement. Even the parts that did not work perfectly were valuable because they showed where the design could be strengthened. For a single-student project, this experience was especially valuable because it required me to take responsibility for every stage of the design, from concept to testing to final presentation.

## Suggestions for Next Version

If this project is developed further in the future, there are many ways it can be improved to make it more useful, reliable, and closer to a real product. The current version works as a student prototype, but the next version could be made stronger in both design and user experience. One of the biggest improvements would be to make the power system better. In the current prototype, many parts are connected, and some features need more stable power than others. In the next version, the power supply should be planned more carefully so that all parts of the system can work smoothly at the same time. This would make the system more reliable and reduce problems during operation.

Another important improvement would be the circuit design. Right now, the project uses separate modules and wiring, which is normal for a prototype. For a better next version, the system could be built on a custom PCB instead of using many separate parts. This would make the design cleaner, safer, and easier to assemble. It would also make the final product look more professional.

The user side of the system could also be improved. The dashboard or control interface should be made easier to use so that a caregiver can understand it quickly without any confusion. Alerts should be clear and easy to notice. For example, the system should clearly show whether the problem is high sound, too much light, or another room condition. A simpler and cleaner interface would make the product more practical for everyday use.

The next version could also be smarter in the way it makes decisions. Now, the system mainly reacts when a sensor value goes above a set limit. A better version could check if the condition lasts for a short time before sending an alert. This would help reduce false alarms. It could also combine more than one sensor reading to make better decisions, which would make the system more accurate.

The enclosure could also be improved. A future version should have a stronger and neater enclosure with better space for wires, sensors, and airflow. This would make the product safer, easier to carry, and more attractive. Since this is a child-room monitoring system, the outside design should also look simple, clean, and safe for home use.

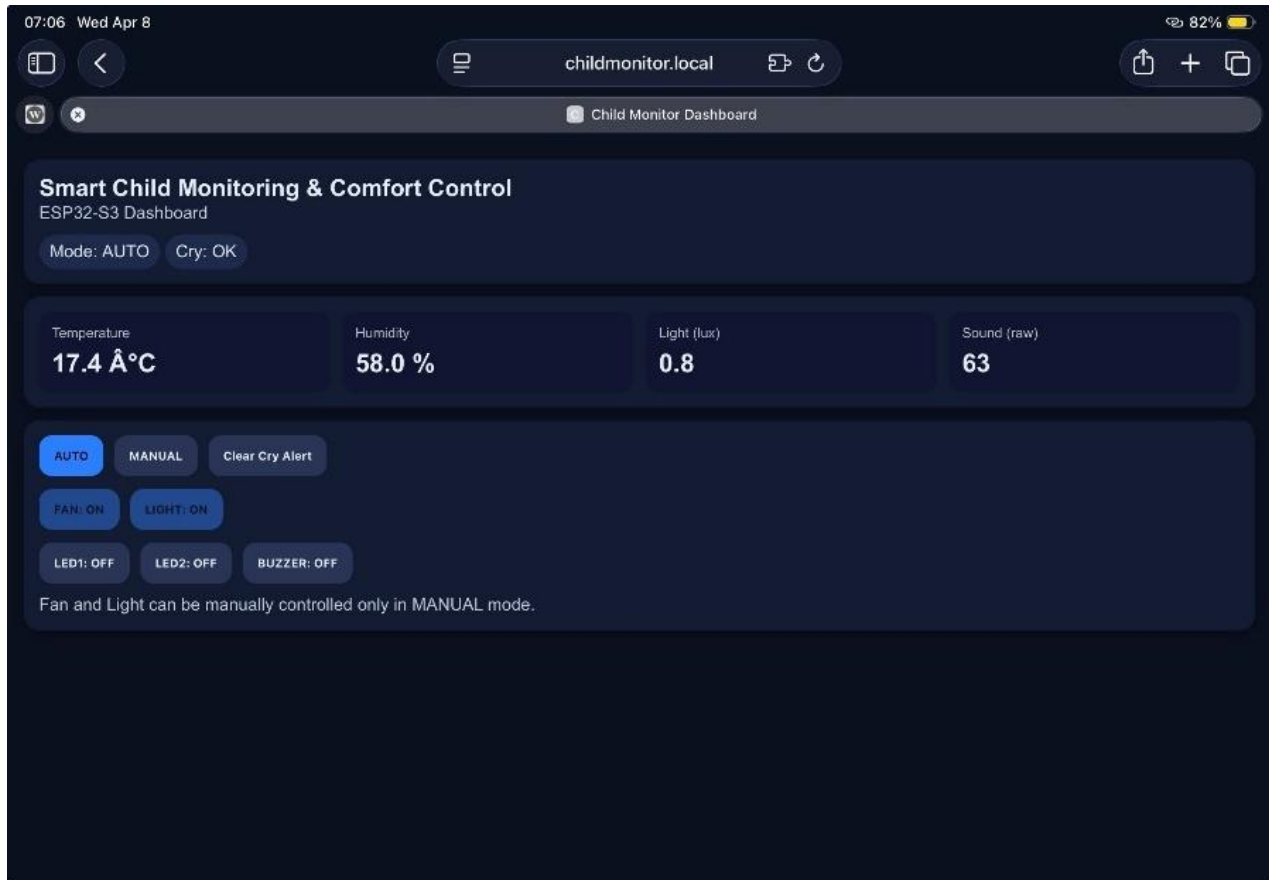
Overall, the next version should focus on better power stability, cleaner circuit design, easier user control, smarter alert logic, and a more polished enclosure. These changes would make the system more dependable, more user-friendly, and more suitable for real-world use.

## References

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- [3] Texas Instruments, “LM2596 SIMPLE SWITCHER® power converter 150-kHz 3-A step-down voltage regulator,” datasheet, Rev. G. [Online]. Available: [TI website](#). [Accessed: Apr. 8, 2026].
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- [5] ROHM Semiconductor, “BH1750FVI digital 16-bit serial output type ambient light sensor IC,” datasheet, Rev. 002, Dec. 6, 2019. [Online]. Available: [ROHM website](#). [Accessed: Apr. 8, 2026].

# Appendices

## Appendix A: Dashboard Screenshot



## Appendix B: ESP32S3-CAM Code

Please find the whole code for Child Monitoring Project by following the link below:

[https://github.com/naman8857/Child-Monitoring-Project-ECE-5000/blob/1a0083dbdb0eb1ddc1638236fe83aafbe2dabf10/Child\\_Monitoring\\_System.ino](https://github.com/naman8857/Child-Monitoring-Project-ECE-5000/blob/1a0083dbdb0eb1ddc1638236fe83aafbe2dabf10/Child_Monitoring_System.ino)